Supplementary Information

Structural regularization and spectral risk in quantum subspace reconstruction

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Online Resource 1: data and reproducibility

The accompanying `Online_Resource_1.zip` contains measurement counts, fitted-model checkpoints, spectra, protocols, source code, analysis tables, and verification scripts. Measurements are classically simulated finite-shot outcomes with ideal sources and gates.

Archive	Primary study	Conditions
`data/L6_mechanism.zip`	Six sites, $U/t = 7$; two steps and four budgets.	160
`data/L8_validation.zip`	Eight sites, $U/t = 7$; two steps and two budgets.	80

Each parameter cell has 20 measurement repetitions. A condition is a parameter–budget–seed combination, not a distinct Hamiltonian. Three additional archives labeled `exploratory` retain earlier latent-model, optimization, and allocation records. They are development records, not further independent validation.

Integrity and reproduction

Run the standard-library audit from the extracted archive root:

```
python3 verify_release.py
```

It checks archive CRCs, container and member hashes, unique records, condition counts, shot budgets, and the eight-site median table. It reads data without executing archived code or retraining models. Per-archive public-export manifests identify the included files and original provenance.

`README.md` gives environment and reproduction commands. Independent analysis code is in `analysis/l6/` and `analysis/l8/`; recorded outputs are in `results/`; original producer modules are in `source/`. Separate producer and analysis dependency lists are supplied.

The full-budget eight-site time-only counts were reconstructed after the original run from recorded sampler settings and seeds. They are labeled as reconstructed in `results/l8/`; they are not new independent trials. Physical-window and periodized errors remain separate metrics.

Export scope

Operational logs, private machine paths, bytecode, and environment reports are omitted. Scientific raw data, models, source modules, and spectra are preserved; a non-scientific path redaction is recorded explicitly. Export ZIP hashes therefore differ from original archive hashes. AI assistance and numerical checks are documented in the manuscript and repository documentation. Repository publication and author submission approval are separate from this supplementary delivery.